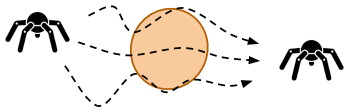
State Constrained Offline RL (SC-ORL)

$$V(x) = \max \sum_{k=0}^{\infty} \gamma^k \cdot r(x_k)$$

$$\text{s.t.} \quad l(x_k) \geq 0 \quad \forall\, k \geq 0$$

Offline Data Collection



$$\mathcal{D} = \{(x_t, a_t, r_t, \ell_t, x_{t+1})\}$$

Solving for SC-ORL

Epigraph Formulation

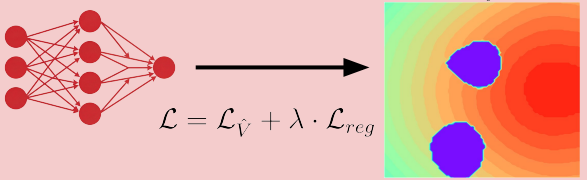
Stage 1:

$$V(x) = \max_{z \in \mathbb{R}} z \qquad \text{s.t.} \quad \hat{V}(x, z) \geq 0$$

Stage 2:

$$\hat{V}(x_t, z_t) = \min\left\{\ell(x_t), \gamma \cdot \hat{V}(x_{t+1}, z_{t+1})\right\}$$

Learning Auxiliary Value Function



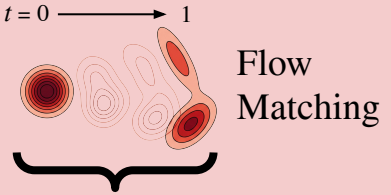
Obtaining Safe & Optimal Policy

Optimal z

Calculate z, using Binary Search

$$z^* = \{\max_{z \in \mathbb{R}} z \mid \hat{V}(x, z) \geq 0\}$$

Learn Optimal Policy



$$\pi^*(a|x) \propto \pi_\beta(a|x) \exp(\alpha \hat{A}(x, z^*(x)))$$